Code No. 9105

100524133019.

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (All Branches) I-Semester (MAIN) Examinations JAN/FEB-2025

ENGINEERING MATHEMATICS – I

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Test the series $\sum_{n=1}^{\infty}\left(\frac{2024}{2025}\right)^n$	2	3	1
b)	State the Cauchy mean value theorem.	2	2	2
c)	Find $f_x(1,-2)$ *and* $f_x(-3,2)$ *for* $f(x,y)=x^2+y^3-2xy$.	2	3	3
d)	Evaluate $\int_2^3\int_1^2 \frac{1}{xy}dy\,dx$	2	3	4
e)	State the Green's theorem.	2	2	5
f)	Find radius of curvature of $y^2=4ax$ *at* $(0,0)$	2	3	
2. a)	Discuss the convergence of the series (i) $\frac{1.2}{3^2.4^2}+\frac{3.4}{5^2.6^2}+\frac{5.6}{7^2.8^2}\ldots\ldots\ldots\ldots$ (ii) $\sum_{n=1}^{\infty}\left(1+\frac{1}{\sqrt{n}}\right)^{-n^{\frac{3}{2}}}x^{2n}$	6	3	1
b)	Test for convergence and absolutely convergence of the series (i) $\sum_{n=1}^{\infty}\frac{\cos n\pi}{n^2+1}$ (ii) $1-\frac{1}{2^3}(1+2)+\frac{1}{3^3}(1+2+3)-\frac{1}{4^3}(1+2+3+4)\ldots\ldots\ldots$	6	3	1
3. a)	Using Lagrange's Mean value theorem show that *if* $a<b$ $\frac{b}{1+b^2}<\tan^{-1}b-\tan^{-1}a<\frac{a}{1+a^2}$ and deduce that $\frac{\pi}{4}+\frac{3}{25}<\tan^{-1}\left(\frac{4}{3}\right)<\frac{\pi}{4}+\frac{1}{6}$	6	4	2

(P.T.O.)

b) Find Taylor's series of $f(x)= 3\cos 2x + 4\sin 4x$, at $x=\frac{\pi}{2}$. 6 3

4. a) Show that the function $f(x,y)=\begin{cases}\dfrac{x^2+y^2}{|x|+|y|}, (x,y)\neq(0,0)\\ 0, (x,y)=(0,0)\end{cases}$ is continuous at (0,0) but its partial derivative $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ doesn't exist at (0,0). 6 3

b) Find the minima and maxima of the function $f(x,y)=21x-12x^2-2y^2+x^3+xy^2$. 6 3

5. a) Calculate area of ellipse by using integration method. 6

b) Evaluate the integrals (i) $\int_0^\infty x^{\frac{5}{2}}e^{-4x^2}dx$ (ii) $\int_0^1 x^5\left(1-x^{\frac{5}{3}}\right)^4 dx$ by using Gamma and Beta functions. 6 3

6. a) If $\tilde{f}= xy^2i + 2x^2yj - 3yz^2k$ then find $div\tilde{f}$ and $curl\tilde{f}$ at a point (1,-1,1). 6 3

b) Find the directional derivative of $\phi(x,y,z)=x^2yz+4xz^2$ at the point (1,-2,-1) in the direction of the normal to the surface $f(x,y,z)=x\log z-y^2$ at (-1,2,-1). 6 3

7. a) Evaluate (i) $\int_0^1\int_y^1 y\sqrt{1+x^3}\,dx\,dy$ (ii) $\int_0^\infty\int_x^\infty \frac{e^{-y}}{y}dy\,dx$ by using change of order of integration method. 6 4

b) If $x=r\sin\theta\cos\phi$, $x=r\sin\theta\sin\phi$, $x=r\cos\theta$ then find $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$. 6 3

****